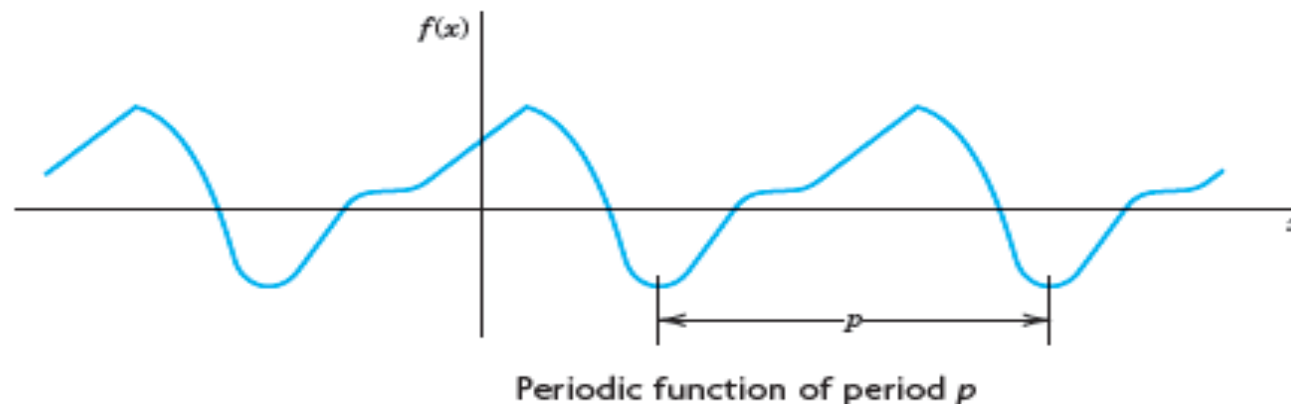


Fourier Series

Fourier series are infinite series that represent periodic functions in terms of cosines and sines. As such, Fourier series are of greatest importance to the engineer and applied mathematician. To define **Fourier series**, we first need some background material. A function $f(x)$ is called a **periodic function** if $f(x)$ is defined for all real x , except possibly at some points, and if there is some positive number p , called a **period** of $f(x)$, such that

$$(1) \qquad f(x + p) = f(x) \qquad \text{for all } x.$$



(The function $f(x) = \tan x$ is a periodic function that is not defined for all real x but undefined for some points (more precisely, countably many points), that is $x = \pm\pi/2, \pm3\pi/2, \dots$)

The graph of a periodic function has the characteristic that it can be obtained by periodic repetition of its graph in any interval of length p

The smallest positive period is often called the *fundamental period*.

Familiar periodic functions are the cosine, sine, tangent, and cotangent. Examples of functions that are not periodic are x , x^2 , x^3 , e^x , $\cosh x$, and $\ln x$, to mention just a few.

If $f(x)$ has period p , it also has the period $2p$ because (1) implies $f(x + 2p) = f([x + p] + p) = f(x + p) = f(x)$, etc.; thus for any integer $n = 1, 2, 3, \dots$,

$$(2) \qquad f(x + np) = f(x) \qquad \text{for all } x.$$

Furthermore if $f(x)$ and $g(x)$ have period p , then $af(x) + bg(x)$ with any constants a and b also has the period p .

Our problem in the first few sections of this chapter will be the representation of various *functions $f(x)$ of period 2π* in terms of the simple functions

$$(3) \quad 1, \quad \cos x, \quad \sin x, \quad \cos 2x, \quad \sin 2x, \cdots, \quad \cos nx, \quad \sin nx, \cdots.$$

All these functions have the period 2π . They form the so-called **trigonometric system**. Figure shows the first few of them (except for the constant 1, which is periodic with any period).

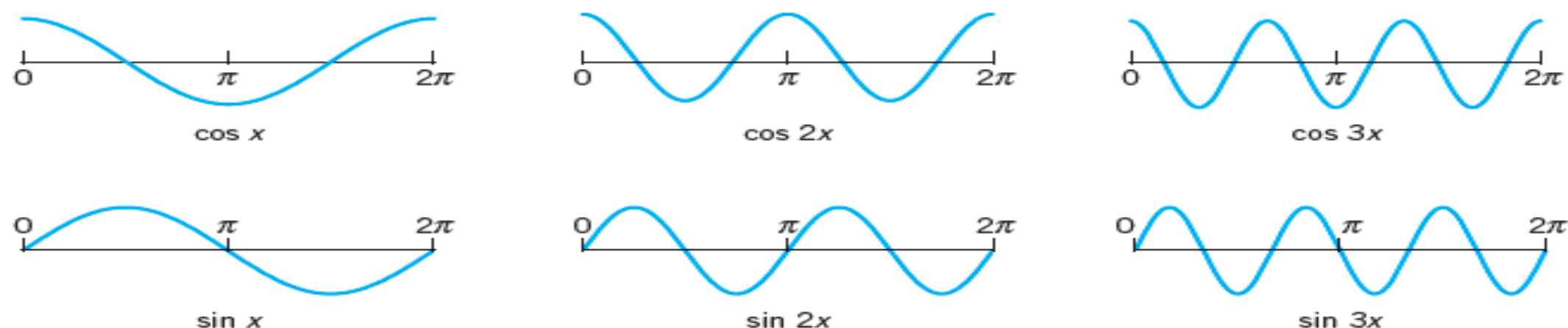


Fig. Cosine and sine functions having the period 2π (the first few members of the trigonometric system (3), except for the constant 1)

The series to be obtained will be a **trigonometric series**, that is, a series of the form

$$\begin{aligned} & a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots \\ (4) \qquad & = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx). \end{aligned}$$

$a_0, a_1, b_1, a_2, b_2, \cdots$ are constants, called the **coefficients** of the series. We see that each term has the period 2π . Hence *if the coefficients are such that the series converges, its sum will be a function of period 2π .*

Expressions such as (4) will occur frequently in Fourier analysis. To compare the expression on the right with that on the left, simply write the terms in the summation. Convergence of one side implies convergence of the other and the sums will be the same.

Now suppose that $f(x)$ is a given function of period 2π and is such that it can be **represented** by a series (4), that is, (4) converges and, moreover, has the sum $f(x)$. Then, using the equality sign, we write

$$(5) \qquad f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

and call (5) the **Fourier series** of $f(x)$.

We shall prove that in this case the coefficients of (5) are the so-called **Fourier coefficients** of $f(x)$, given by the **Euler formulas**

$$\begin{aligned} (0) \quad & a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx \\ (6) \quad (a) \quad & a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \quad n = 1, 2, \dots \\ (b) \quad & b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \quad n = 1, 2, \dots \end{aligned}$$

Find the Fourier coefficients of the periodic function $f(x)$. The formula is

$$(7) \quad f(x) = \begin{cases} -k & \text{if } -\pi < x < 0 \\ k & \text{if } 0 < x < \pi \end{cases} \quad \text{and} \quad f(x + 2\pi) = f(x).$$

$$a_0 = 0.$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \cos nx \, dx + \int_0^{\pi} k \cos nx \, dx \right] \\ &= \frac{1}{\pi} \left[-k \frac{\sin nx}{n} \Big|_{-\pi}^0 + k \frac{\sin nx}{n} \Big|_0^{\pi} \right] = 0 \end{aligned}$$

because $\sin nx = 0$ at $-\pi, 0$, and π for all $n = 1, 2, \dots$. We see that all these cosine coefficients are zero. That is, the Fourier series of (7) has no cosine terms, just sine terms, it is a **Fourier sine series** with coefficients $b_1, b_2, \dots$ obtained from (6b);

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \sin nx \, dx + \int_0^{\pi} k \sin nx \, dx \right] \\
 &= \frac{1}{\pi} \left[k \frac{\cos nx}{n} \Big|_{-\pi}^0 - k \frac{\cos nx}{n} \Big|_0^{\pi} \right].
 \end{aligned}$$

Since $\cos(-\alpha) = \cos \alpha$ and $\cos 0 = 1$, this yields

$$b_n = \frac{k}{n\pi} [\cos 0 - \cos(-n\pi) - \cos n\pi + \cos 0] = \frac{2k}{n\pi} (1 - \cos n\pi).$$

Now, $\cos \pi = -1$, $\cos 2\pi = 1$, $\cos 3\pi = -1$, etc.; in general,

$$\cos n\pi = \begin{cases} -1 & \text{for odd } n, \\ 1 & \text{for even } n, \end{cases} \quad \text{and thus} \quad 1 - \cos n\pi = \begin{cases} 2 & \text{for odd } n, \\ 0 & \text{for even } n. \end{cases}$$

Hence the Fourier coefficients b_n of our function are

$$b_1 = \frac{4k}{\pi}, \quad b_2 = 0, \quad b_3 = \frac{4k}{3\pi}, \quad b_4 = 0, \quad b_5 = \frac{4k}{5\pi}, \dots$$

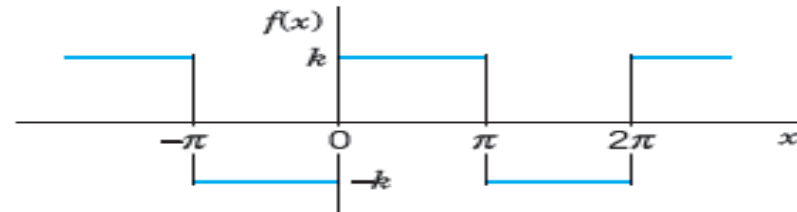


Fig. Given function $f(x)$ (Periodic rectangular wave)

Since the a_n are zero, the Fourier series of $f(x)$ is

$$(8) \quad \frac{4k}{\pi} (\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots).$$

Orthogonality of the Trigonometric System (3)

The trigonometric system (3) is orthogonal on the interval $-\pi \leq x \leq \pi$ (hence also on $0 \leq x \leq 2\pi$ or any other interval of length 2π because of periodicity); that is, the integral of the product of any two functions in (3) over that interval is 0, so that for any integers n and m ,

$$(a) \quad \int_{-\pi}^{\pi} \cos nx \cos mx \, dx = 0 \quad (n \neq m)$$

$$(9) \quad (b) \quad \int_{-\pi}^{\pi} \sin nx \sin mx \, dx = 0 \quad (n \neq m)$$

$$(c) \quad \int_{-\pi}^{\pi} \sin nx \cos mx \, dx = 0 \quad (n \neq m \text{ or } n = m).$$

PROOF

$$\int_{-\pi}^{\pi} \cos nx \cos mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \cos (n + m)x \, dx + \frac{1}{2} \int_{-\pi}^{\pi} \cos (n - m)x \, dx$$
$$\int_{-\pi}^{\pi} \sin nx \sin mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \cos (n - m)x \, dx - \frac{1}{2} \int_{-\pi}^{\pi} \cos (n + m)x \, dx.$$

Since $m \neq n$ (integer!), the integrals on the right are all 0. Similarly, in (9c), for all integer m and n

$$\int_{-\pi}^{\pi} \sin nx \cos mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \sin (n + m)x \, dx + \frac{1}{2} \int_{-\pi}^{\pi} \sin (n - m)x \, dx = 0 + 0.$$

Application of Theorem 1 to the Fourier Series (5)

We prove (6.0). Integrating on both sides of (5) from $-\pi$ to π , we get

$$\int_{-\pi}^{\pi} f(x) \, dx = \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right] dx.$$

We now assume that termwise integration is allowed.

Then we obtain

$$\int_{-\pi}^{\pi} f(x) \, dx = a_0 \int_{-\pi}^{\pi} dx + \sum_{n=1}^{\infty} \left(a_n \int_{-\pi}^{\pi} \cos nx \, dx + b_n \int_{-\pi}^{\pi} \sin nx \, dx \right).$$

The first term on the right equals $2\pi a_0$. Integration shows that all the other integrals are 0. Hence division by 2π gives (6.0).

We prove (6a). Multiplying (5) on both sides by $\cos mx$ with any *fixed* positive integer m and integrating from $-\pi$ to π , we have

$$(10) \quad \int_{-\pi}^{\pi} f(x) \cos mx \, dx = \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right] \cos mx \, dx.$$

We now integrate term by term. Then on the right we obtain an integral of $a_0 \cos mx$, which is 0; an integral of $a_n \cos nx \cos mx$, which is $a_m \pi$ for $n = m$ and 0 for $n \neq m$ by (9a); and an integral of $b_n \sin nx \cos mx$, which is 0 for all n and m by (9c). Hence the right side of (10) equals $a_m \pi$. Division by π gives (6a) (with m instead of n).

We finally prove (6b). Multiplying (5) on both sides by $\sin mx$ with any *fixed* positive integer m and integrating from $-\pi$ to π , we get

$$(11) \quad \int_{-\pi}^{\pi} f(x) \sin mx \, dx = \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right] \sin mx \, dx.$$

Integrating term by term, we obtain on the right an integral of $a_0 \sin mx$, which is 0; an integral of $a_n \cos nx \sin mx$, which is 0 by (9c); and an integral of $b_n \sin nx \sin mx$, which is $b_m \pi$ if $n = m$ and 0 if $n \neq m$, by (9b). This implies (6b) (with n denoted by m). This completes the proof of the Euler formulas (6) for the Fourier coefficients.

THEOREM 2

Representation by a Fourier Series

Let $f(x)$ be periodic with period 2π and piecewise continuous in the interval $-\pi \leq x \leq \pi$. Furthermore, let $f(x)$ have a left-hand derivative and a right-hand derivative at each point of that interval. Then the Fourier series (5) of $f(x)$ [with coefficients (6)] converges. Its sum is $f(x)$, except at points x_0 where $f(x)$ is discontinuous. There the sum of the series is the average of the left- and right-hand limits of $f(x)$ at x_0 .

1. From Period 2π to Any Period $p = 2L$

Clearly, periodic functions in applications may have any period, not just 2π as in the last section (chosen to have simple formulas). The notation $p = 2L$ for the period is practical because L will be a length of a violin string, of a rod in heat conduction in $\frac{1}{2}$ of a cycle, and so on.

The transition from period 2π to be period $p = 2L$ is effected by a suitable change of scale, as follows. Let $f(x)$ have period $p = 2L$. Then we can introduce a new variable v such that $f(x)$, as a function of v , has period 2π . If we set

$$(1) \quad (a) \quad x = \frac{p}{2\pi} v, \quad \text{so that} \quad (b) \quad v = \frac{2\pi}{p} x = \frac{\pi}{L} x$$

then $v = \pm\pi$ corresponds to $x = \pm L$. This means that f , as a function of v , has period 2π and, therefore, a Fourier series of the form

$$(2) \quad f(x) = f\left(\frac{L}{\pi} v\right) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nv + b_n \sin nv)$$

with coefficients obtained from (6) in the last section

$$(3) \quad \begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) dv, & a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) \cos nv \, dv, \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) \sin nv \, dv. \end{aligned}$$

We could use these formulas directly, but the change to x simplifies calculations. Since

$$(4) \quad v = \frac{\pi}{L}x, \quad \text{we have} \quad dv = \frac{\pi}{L} dx$$

and we integrate over x from $-L$ to L . Consequently, we obtain for a function $f(x)$ of period $2L$ the Fourier series

$$(5) \quad f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{L} x + b_n \sin \frac{n\pi}{L} x \right)$$

with the **Fourier coefficients** of $f(x)$ given by the **Euler formulas** (π/L in dx cancels $1/\pi$ in (3))

$$(6) \quad \begin{aligned} (0) \quad a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx \\ (a) \quad a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx & n = 1, 2, \dots \\ (b) \quad b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx & n = 1, 2, \dots \end{aligned}$$

we continue to call (5) with any coefficients a **trigonometric series**. And we can integrate from 0 to $2L$ or over any other interval of length $p = 2L$.

Find the Fourier series of the function

$$f(x) = \begin{cases} 0 & \text{if } -2 < x < -1 \\ k & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x < 2 \end{cases} \quad p = 2L = 4, \quad L = 2.$$

Solution. From (6.0) we obtain $a_0 = k/2$ (verify!). From (6a) we obtain

$$a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi x}{2} dx = \frac{1}{2} \int_{-1}^1 k \cos \frac{n\pi x}{2} dx = \frac{2k}{n\pi} \sin \frac{n\pi}{2}.$$

Thus $a_n = 0$ if n is even and

$$a_n = 2k/n\pi \quad \text{if } n = 1, 5, 9, \dots, \quad a_n = -2k/n\pi \quad \text{if } n = 3, 7, 11, \dots.$$

From (6b) we find that $b_n = 0$ for $n = 1, 2, \dots$. Hence the Fourier series is a **Fourier cosine series** (that is, it has no sine terms)

$$f(x) = \frac{k}{2} + \frac{2k}{\pi} \left(\cos \frac{\pi}{2} x - \frac{1}{3} \cos \frac{3\pi}{2} x + \frac{1}{5} \cos \frac{5\pi}{2} x - + \dots \right).$$

2. Simplifications: Even and Odd Functions

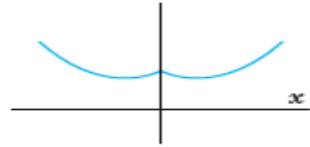


Fig.
Even function

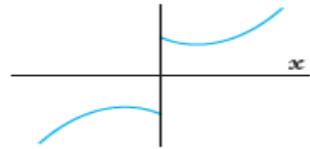


Fig.
Odd function

If $f(x)$ is an **even function**, that is, $f(-x) = f(x)$, its Fourier series (5) reduces to a **Fourier cosine series**

$$(5^*) \quad f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{L} x \quad (f \text{ even})$$

with coefficients (note: integration from 0 to L only!)

$$(6^*) \quad a_0 = \frac{1}{L} \int_0^L f(x) dx, \quad a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx, \quad n = 1, 2, \dots$$

If $f(x)$ is an **odd function**, that is, $f(-x) = -f(x)$, its Fourier series (5) reduces to a **Fourier sine series**

$$(5^{**}) \quad f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{L} x \quad (f \text{ odd})$$

with coefficients

$$(6^{**}) \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx.$$

These formulas follow from (5) and (6) by remembering from calculus that the definite integral gives the net area (= area above the axis minus area below the axis) under the curve of a function between the limits of integration. This implies

$$(7) \quad \begin{aligned} \text{(a)} \quad \int_{-L}^L g(x) dx &= 2 \int_0^L g(x) dx && \text{for even } g \\ \text{(b)} \quad \int_{-L}^L h(x) dx &= 0 && \text{for odd } h \end{aligned}$$

Formula (7b) implies the reduction to the cosine series (even f makes $f(x) \sin (n\pi x/L)$ odd since $\sin$ is odd) and to the sine series (odd f makes $f(x) \cos (n\pi x/L)$ odd since $\cos$ is even). Similarly, (7a) reduces the integrals in (6*) and (6**) to integrals from 0 to L . These reductions are obvious from the graphs of an even and an odd function.

Summary

Even Function of Period 2π . If f is even and $L = \pi$, then

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

with coefficients

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx, \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx, \quad n = 1, 2, \dots$$

Odd Function of Period 2π . If f is odd and $L = \pi$, then

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

with coefficients

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx, \quad n = 1, 2, \dots$$